

Sequences and series

Arithmetic and geometric progressions

YOUR AIM TODAY

Find terms and sums, then decide which model applies.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

The ideas you need

A progression follows a fixed rule from one term to the next.

- AP: common difference d ; n th term $a+(n-1)d$.
- AP sum: $n/2[2a+(n-1)d]$.
- GP: common ratio r ; n th term $ar^{(n-1)}$.
- GP sum: $a(1-r^n)/(1-r)$.
- Infinite GP sum $a/(1-r)$ only when $|r|<1$.

MEMORY RULE

Subtract neighbouring terms for an AP; divide them for a GP.

AP

WORKED STEPS

3, 7, 11, ... has $a=3, d=4$. Term 20 = $3+19(4)=79$.

GP

WORKED STEPS

5, 10, 20, ... has $a=5, r=2$. Sum first 4 = $5(1-16)/(1-2)=75$.

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Find term 15 of 4,7,10,...	
2. Find sum first 20 terms of 2,6,10,...	
3. Find term 8 of 3,6,12,...	
4. Find sum first 6 terms of 2,6,18,...	
5. Find infinite sum $12+6+3+\dots$	
6. In an AP, term 3=11 and term 8=31. Find a and d.	

STOP

Do not continue until all six questions have an attempt.

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	$a=4, d=3: 4+14(3)=46.$
2	$n/2[2a+(n-1)d]=10[4+76]=800.$
3	$3(2^7)=384.$
4	$2(1-3^6)/(1-3)=728.$
5	$12/(1-1/2)=24.$
6	$a+2d=11, a+7d=31, \text{ so } d=4 \text{ and } a=3.$

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.